



DATABASE-SYSTEMS

Lecture - 3

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Database Development Process

- Basic purposed of the database development is to satisfy the user requirements
- Sometimes user interviews are also conducted to find the user needs
- The two major strategies to develop database are following:

**TOP DOWN
DEVELOPMENT**

**BOTTOM UP
DEVELOPMENT**

Database Development Process

TOP DOWN – DEVELOPMENT

- This strategies starts with **general issues moving towards specific issues**.
- Frist of all **general goals** of an organization are **identified**.
- **Requirements** are identified to **satisfy** those **goals**.
- This study gives **abstract data-models** that helps to define **specific data-models**.
- Then **high level** data-models are **transformed** into the **low level** models.

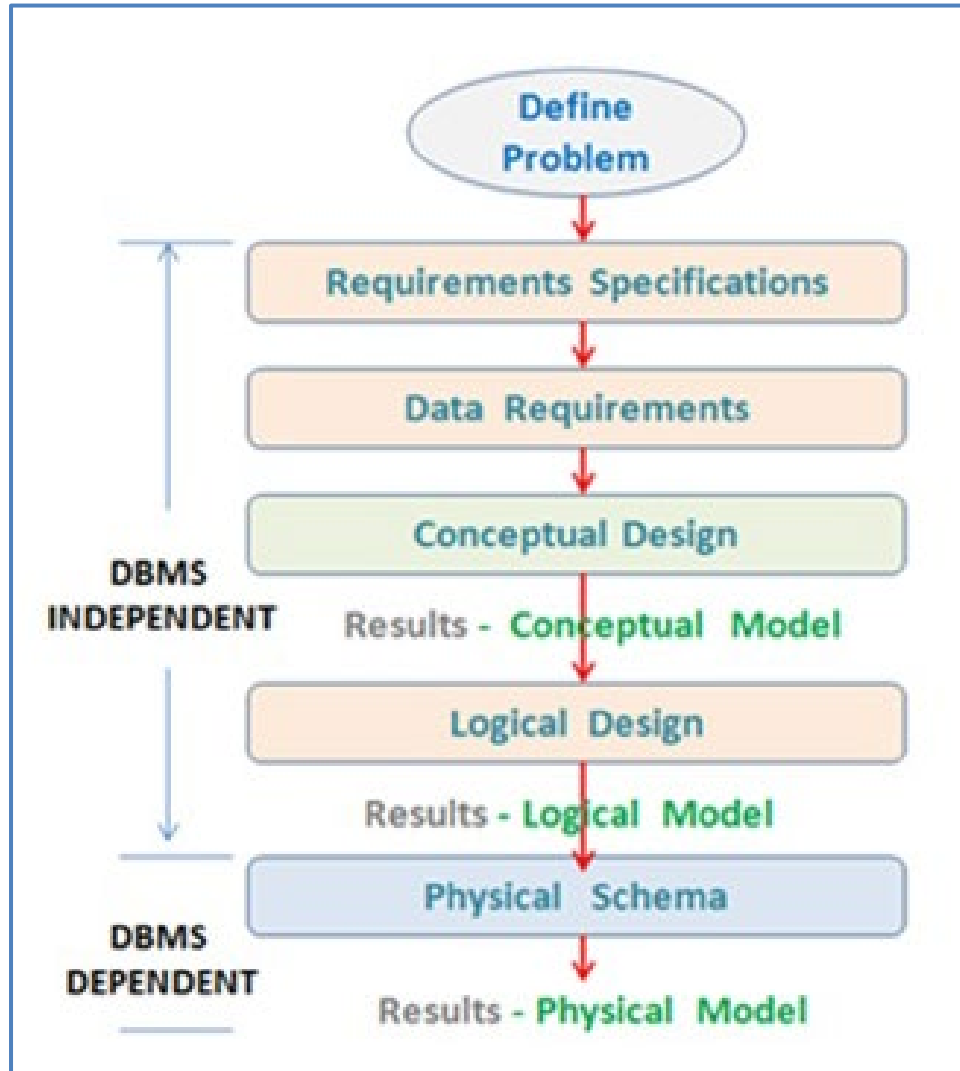
BOTTOM UP – DEVELOPMENT

- This model starts with **specific issues moving towards general issues**.
- Frist of all **specific goals** of an organization are **identified**.
- **Requirements** are **collected** about the **specific modules** of database systems.
- Then then specific modules are **developed & combined** to create **whole** system.
- **Data-Models** for **each** module is **developed separately**.



Database Development Process

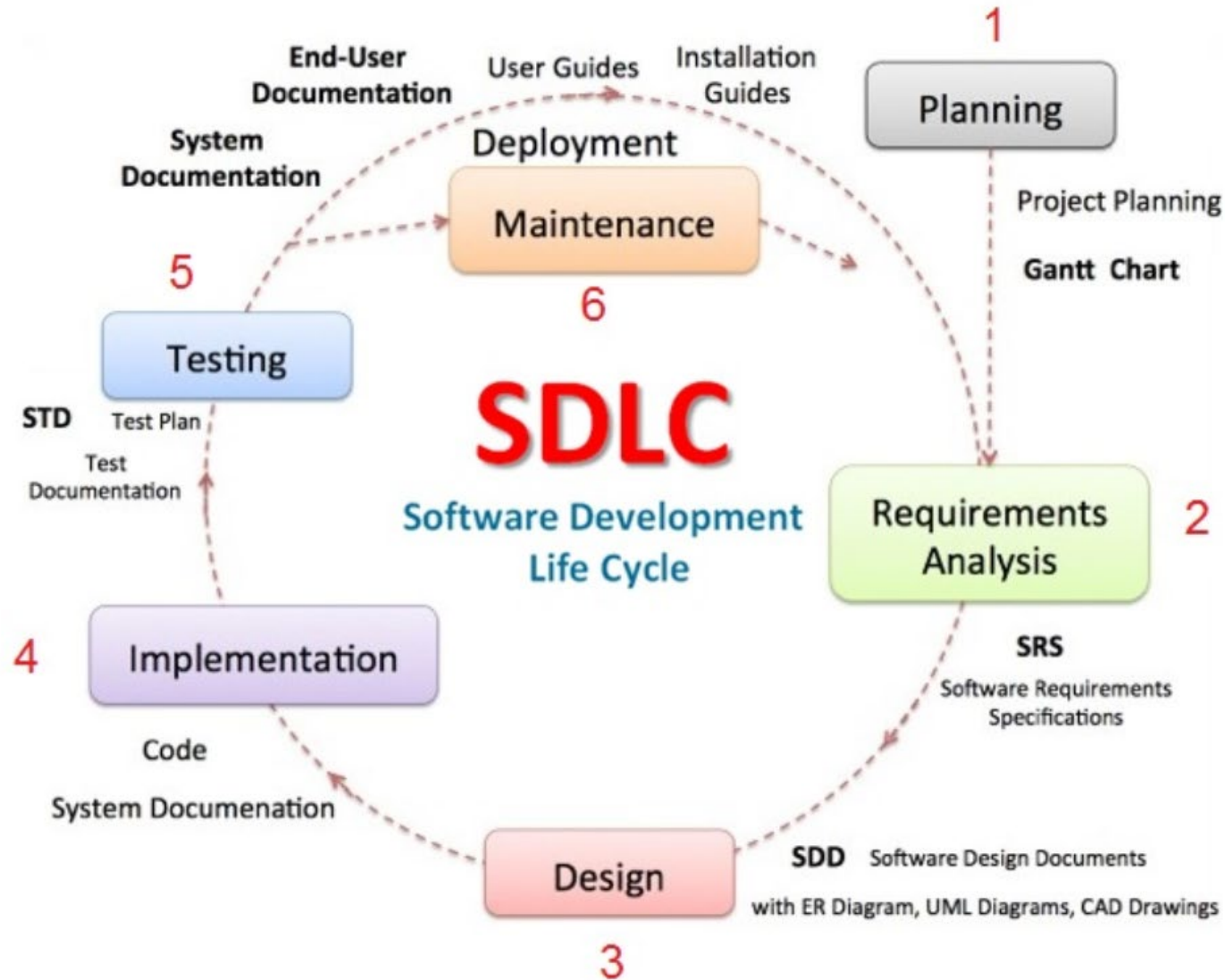
System Development Life Cycle for Database



1. Define Database design Objectives
2. Database designers collect the user database requirements by elicitation .
3. Database designers produce conceptual data model then expanded to Logical data model then expanded to physical data model .
4. Database designers construct logical design of the database with tables , Defining Relations , DB Constraints by
5. Database designers then Refine & Normalize the database to minimize the data redundancy
6. Physical implementation of database is managed by the DBMS .

Database Development Process

System Development Life Cycle for Database



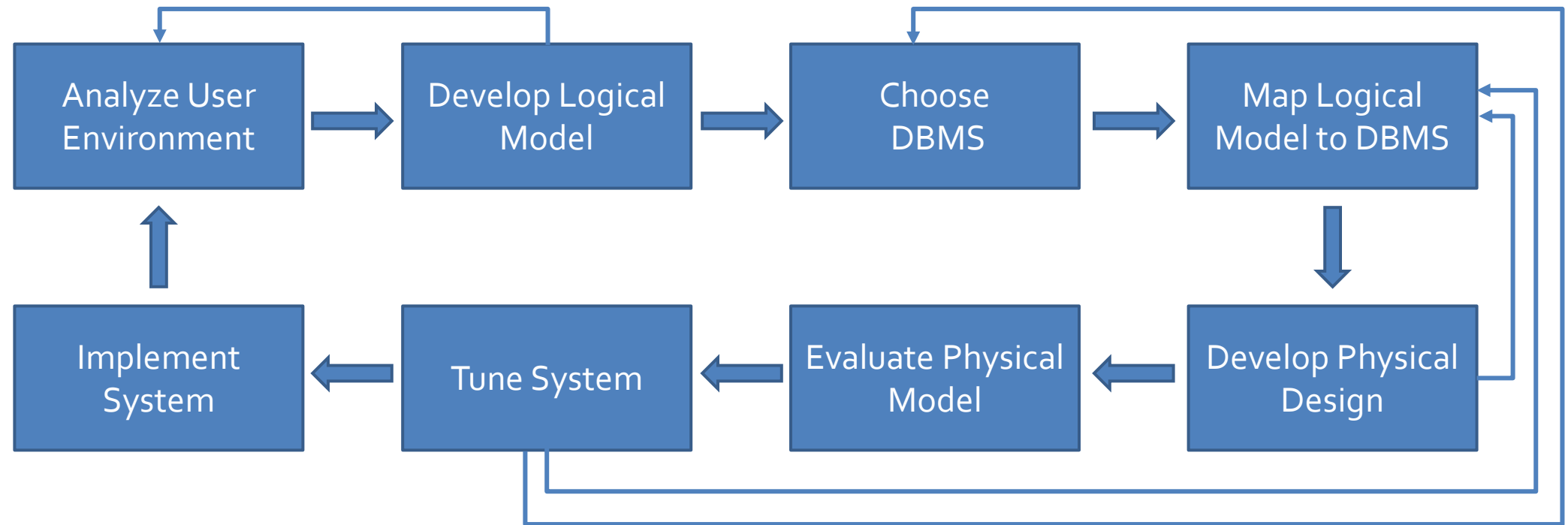
Staged Database Design Approach

Stages of Staged Approach (It is a type of top down approach)

- | | |
|---|----------------------------|
| 1 | Analyze User Environment |
| 2 | Develop Logical Data Model |
| 3 | Choose a DBMS |
| 4 | Map Logical Model to DBMS |
| 5 | Develop Physical Model |
| 6 | Evaluate Physical Model |
| 7 | Perform Tuning |
| 8 | Implement Physical Model |



Staged Database Design Approach



Database Administration

Database Administrator (DBA) is the main person responsible for following

1	Preliminary Database Planning	9	Writing & Maintaining Documentation
2	Identifying User Requirements	10	Developing & Enforcing Data Standards
3	Developing and Maintaining Data Dictionary	11	Developing Operating Procedures
4	Choosing A DBMS	12	Training Users
5	Developing Physical Model	13	Helping Database Users
6	Creating & Loading Databases	14	Defining Backup & Recovery Procedures
7	Developing User Views	15	Monitoring Performance
8	Implement Physical Model	16	Tuning & Reorganization

